QP CODE:25901680

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

REGULAR/IMPROVEMENT/REAPPEARANCEEXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCFTQ100 - FOOD ADDITIVES

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Part A

Short Answer Questions

Answer any 5 questions

Each question carries 1 marks

1. **In which fruits and vegetables is lycopene predominantly found? [U] / [CO5]**
   Tomatoes and tomato based products; also watermelon, pink guava, papaya
2. **Which metal ion can catalyze oxidation in food products? [K] / [CO4]**
   Iron (Fe^{2+}/Fe^{3+}); copper (Cu^{2+})
3. **What is the primary use of aspartame in food products? [U] / [CO5]**
   Artificial sweetener used as a low calorie sugar substitute
4. **What makes honey different from other natural sweeteners? [K] / [CO5]**
   Contains enzymes, antioxidants, and antimicrobial compounds; high fructose content
5. **What is the primary function of hydrocolloids in food? [U] / [CO5]**
   Thickening, gelling, stabilizing, and improving texture
6. **What is one method commonly used for flavour encapsulation? [K] / [CO5]**
   Spray drying
7. **What is the function of bulking agents in food? [U] / [CO3]**
   Provide volume and texture with minimal caloric contribution

Part B

Short Essay Questions

*Answer any **5** questions*

*Each question carries **4** marks*

8. **How do preservatives help extend the shelf life of food? [K] / [CO1]**

   Inhibit growth of microorganisms such as bacteria, yeasts, and molds

   Slow down enzymatic spoilage and oxidation reactions

   Maintain food quality, safety, and sensory properties

   Examples include salt, sugar, organic acids, and chemical preservatives

9. **What is the definition of food preservatives and what is their purpose in food? [K] / [CO3]**

   Food preservatives are substances added to food to prevent spoilage

   Their purpose is to extend shelf life by inhibiting microbial growth and chemical deterioration

   Help retain freshness, texture, color, and nutritional value

10. **How do fungicidal agents function in preventing fungal growth? [U] / [CO2]**

    Disrupt fungal cell membrane integrity and permeability

    Inhibit enzyme systems essential for fungal metabolism

    Interfere with spore germination and mycelial growth

    Examples include sorbates and propionates

11. **Explain the role of additives in producing sugar-free or low-fat alternatives. [K] / [CO5]**

    Artificial sweeteners replace sugar to provide sweetness with fewer calories

    Bulking agents maintain texture and volume in absence of sugar

    Fat replacers mimic mouthfeel and consistency of fats

    Stabilizers and emulsifiers improve product structure and acceptability

12. **What is the function of leavening agents in baking? [K] / [CO5]**

    Produce carbon dioxide gas during baking

    Cause dough or batter to rise and become light and porous

    Improve texture, volume, and softness of baked products

    Examples include baking powder, baking soda, and yeast

13. **Discuss the different types of emulsifiers used in food products. [K] / [CO5]**

    Natural emulsifiers: lecithin from egg yolk or soy

    Synthetic emulsifiers: mono- and diglycerides

    Protein-based emulsifiers: casein, whey proteins

    Polysaccharide-based emulsifiers: gums and starch derivatives

14. **What are the methods used for extracting natural colorants from plants? [K] / [CO5]**

    Solvent extraction using water, ethanol, or acetone

    Steam distillation for volatile pigments

    Cold pressing for pigments in peels

    Enzyme-assisted extraction to improve yield

    Supercritical fluid extraction for high-purity pigments

Part C

Long Essay Questions

*Answer any **1** questions*

*Each question carries **10** marks*

15. **Discuss the role of hydrocolloids, leavening agents and pH control agents in food industry. [K] / [CO5]**

    Hydrocolloids act as thickening, gelling, and stabilizing agents; improve texture, viscosity, and moisture retention

    Leavening agents release carbon dioxide, making baked products soft, porous, and increased in volume

    pH control agents regulate acidity or alkalinity; enhance flavor, inhibit microbial growth, and improve stability

16. **Describe the function of buffering salts and anticaking agents in food processing. How do they help maintain the stability and quality of food products? [K] / [CO5]**

    Buffering salts maintain a stable pH, preventing undesirable changes in flavor, texture, and microbial growth

    Help maintain product consistency during processing and storage

Anticaking agents prevent lump formation in powdered foods by reducing moisture absorption

Improve flow properties, handling, and shelf stability of products

17. **How do synthetic antioxidants prevent food oxidation, and what are the pros and cons of using them in food processing? [K] / [CO5]**

Synthetic antioxidants (e.g., BHA, BHT) donate hydrogen atoms to free radicals, interrupting oxidation chain reactions

Prevent rancidity, off-flavors, and nutrient loss in foods

Highly effective at low concentrations

Extend shelf life and maintain quality

Cost-effective and stable

Disadvantages

Potential health concerns at high intake levels

Consumer preference shifting toward natural additives

Regulatory restrictions in some regions